Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Integrated Pest, Disease, and Nematode Management

1.1 Major Sucking Insect Pests: Biology, Injury Grades, and Vector Roles

- **Cotton Leafhopper / Jassid (*Amrasca biguttula biguttula*):**
  - *Taxonomy and Life Cycle:* Belongs to Cicadellidae (Hemiptera). Polyphagous pest with piercing and sucking mouthparts. Nymphal development completes in 5 to 16 days through five instars, while adult longevity ranges from 13 to 23 days. Females lay 15 to 30 pale-green eggs singly inside leaf midribs and major veins. Completes 11 to 15 overlapping generations annually across India due to continuous alternate host availability (okra, brinjal, potato, sunflower, castor).
  - *Damage Mechanism:* Nymphs and adults concentrate on the lower abaxial leaf surface to extract cell sap while simultaneously injecting toxic salivary secretions.
  - *Standard Injury Grading Scale:*
    - **Grade-I:** Entire foliage completely green, free from leaf crinkling or curling, with no chlorosis.
    - **Grade-II:** Crinkling and downward/upward curling of lower leaves accompanied by marginal leaf yellowing.
    - **Grade-III:** Severe crinkling, cupping, and curling across upper and middle canopy leaves, accompanied by stunted plant growth.
    - **Grade-IV:** Extreme curling, bronzing, browning, and complete drying of leaf margins ("hopperburn"), culminating in premature foliar desiccation and leaf drop.
- **Cotton Whitefly (*Bemisia tabaci*):**
  - *Taxonomy and Life Cycle:* Belongs to Aleyrodidae (Hemiptera). Tiny white-winged adults (1.0 mm) lay stalked, yellow, sub-elliptical eggs on lower leaves. Eggs hatch into mobile crawlers that settle into scale-like sessile nymphs (2nd, 3rd, and 4th puparial instars). Life cycle completes in 14 to 24 days under warm, humid weather, producing 10 to 12 generations annually. Population peaks occur in North India during two major windows: mid-July (31st–32nd SMW) and mid-September (37th–38th SMW).

  - *Sap Depletion & Sooty Mould Development:* Nymphs and adults suck phloem sap, causing leaf chlorosis, plant weakness, and square/boll shedding. Insects excrete honeydew that coats foliage and green bolls, fostering black sooty mould (*Capnodium* spp.) that inhibits photosynthesis and turns harvested lint sticky, black, and unmarketable.
  - *Disease Vector Capability:* Serves as the exclusive biological vector transmitting Cotton Leaf Curl Virus (CLCuV) in a persistent, circulative manner.

- **Cotton Thrips (*Thrips tabaci*):**
  - *Taxonomy and Damage Profile:* Belongs to Thripidae (Thysanoptera). Minute, slender, fringed-winged insects. Adults and nymphs use punch-and-suck mouthparts to lacerate upper and lower epidermal cells, lapping up emerging cell sap.
  - *Diagnostic Field Symptoms:* Leaves display characteristic silvery-white patches, streaks, or bronzed undersides. Affected young leaves curl upward, become thick, leathery, and brittle, while terminal buds dry up, leading to terminal growth suppression and severe square shedding. Serves as a mechanical and biological vector for Tobacco Streak Virus (TSV).

- **Cotton Aphid (*Aphis gossypii*):**
  - *Damage Symptoms:* Small, soft-bodied, yellowish-green to dark black insects displaying apterous (wingless) and alate (winged) forms. Succulent shoots, buds, and lower leaf surfaces are covered by dense colonies. Sap removal causes downward cupping, leaf crumpling, stunting, and wilting. Excretes honeydew leading to sooty mould and lint staining.

- **Phenacoccus Mealybug (*Phenacoccus solenopsis*):**
  - *Damage Symptoms:* Highly invasive pseudococcid sucking pest covered with a white waxy mealy secretion. Nymphs and wingless adult females suck sap from main stems, petioles, leaf veins, and boll pedicels. Feeding causes stem malformation, severe leaf crinkling, small boll size, failure of boll opening, and total plant death in circular field patches.

- **Mirid Bugs (*Creontiades biseratense* and *Campylomma livida*):**
  - *Damage Symptoms:* Slender green/brown phytophagous bugs piercing squares, pinhead buds, and young green bolls. Feeding produces pinhole punctures, dark brown necrotic spots, square drop, and deformed, asymmetric "parrot-beaked" green bolls with stained lint.

1.2 Bollworm Complex and Defoliating Caterpillars

- **Pink Bollworm (*Pectinophora gossypiella*):**
  - *Taxonomy and Monophagous Habits:* Belongs to Gelechiidae (Lepidoptera). Functionally monophagous pest attacking cultivated cotton species (*G. hirsutum*, *G. arboreum*, *G. herbaceum*, *G. barbadense*). Life cycle completes in 25 to 30 days under warm conditions.

  - *Diapause and Carryover:* In North and Central India, the final late-season generation enters long winter diapause as fully mature larvae inside unopened bolls, double-joined seeds, gin trash, and stacked cotton stalks. Diapause duration lasts between 31 and 157 days until spring warmth and high humidity trigger pupation.
  - *Diagnostic Symptoms:*
    * **Rosette Flowers:** Newly hatched 1st instar larvae bore into unopened floral buds, webbing the inner margins of petals together with fine silk, producing non-opening, twisted "rosette flowers".
    * **Invisible Carpel Entry:** Larvae bore directly into young green bolls (15–28 days old) through tiny entry holes that heal quickly, leaving no visible external entry marks.
    * **Internal Lock Destruction:** Larvae tunnel inside carpel walls, devouring developing seed kernels and lint fibers, filling cavities with reddish-brown frass, preventing normal boll opening, staining fiber yellow/brown, and reducing seed oil and germination.

- **American Bollworm (*Helicoverpa armigera*):**
  - *Taxonomy and Polyphagous Nature:* Belongs to Noctuidae (Lepidoptera). Highly destructive polyphagous pest. Female moths lay spherical, sculptured, yellowish-white eggs singly on tender leaves, squares, and terminal shoots.
  - *Damage Symptoms:* Caterpillars feed on squares, flowers, and green bolls by thrusting the anterior body inside the fruiting organ while leaving the posterior half exposed. Infested squares exhibit yellowing and outward flaring of epicalyx bracteoles ("flared-up squares"), followed by premature dropping. Bolls show large, clean, circular entrance holes surrounded by dark frass deposits.

- **Spiny / Spotted Bollworms (*Earias insulana* and *Earias vittella*):**
  - *Taxonomy and Life Cycle:* Belong to Nolidae (Lepidoptera). Females lay crown-shaped, bluish-green eggs singly on tender buds and terminal shoots.
  - *Damage Symptoms:* During early vegetative growth, larvae bore into central terminal shoots, causing terminal drying, collapse, and wilting ("dead hearts"). During flowering, larvae bore into squares and green bolls, producing flared-up squares, plugged entrance holes oozing wet brownish excreta, and hollowed bolls.

- **Tobacco Caterpillar (*Spodoptera litura*):**
  - *Taxonomy and Feeding Habits:* Polyphagous noctuid pest. Female moths lay golden-brown hair-covered egg masses (200–300 eggs) on lower leaf surfaces.
  - *Damage Symptoms:* 1st and 2nd instar larvae feed gregariously on lower epidermal layers, skeletonizing foliage into paper-white membranous patches. Older larvae scatter across the canopy, devouring entire leaf laminas until only main veins remain (sieve-like foliage appearance), and bore into squares and green bolls.

- **Dusky Cotton Bug (*Oxycarenus hyalinipennis*):**

  - *Damage Symptoms:* Lygaeid bug. Nymphs and adults congregate inside open bolls to suck sap from mature seeds, lowering seed weight, oil content, and germination. Bugs crushed during mechanical ginning permanently stain lint brown/red.

1.3 Minor, Subterranean, and Emerging Pests

- **Cotton Stem Weevil (*Pempherulus affinis*):** Curculionid pest. Grubs bore into the sub-epidermal stem region near ground level, forming galls or gall-like swellings. Affected young plants exhibit stem breakage, wilting, and lodging during high wind events.
- **Cotton Leaf Folder / Semi-loopers (*Sylepta derogata* and *Anomis flava*):** Caterpillars roll leaves into characteristic trumpet-like cylinders, binding edges with silk threads and feeding inside on green leaf parenchyma, causing defoliation.
- **Two-Spotted Spider Mite (*Tetranychus urticae*):** Phytophagous tetranychid mites colony on lower leaf surfaces under warm, dry conditions. Feeding produces reddish-brown or bronzed interveinal patches, webbing, and premature leaf drop.
- **Subterranean Termites (*Odontotermes obesus*):** Attack roots and sub-surface main stem tissues in light, rainfed soils, destroying vascular channels and causing plant wilting and drying.

1.4 Bacterial and Fungal Diseases of Cotton

- **Bacterial Blight / Angular Leaf Spot / Black Arm / Boll Rot (*Xanthomonas citri* pv. *malvacearum*):**
  - *Causal Agent and Transmission:* Caused by the gram-negative bacterium *X. citri* pv. *malvacearum* (syn. *X. campestris* pv. *malvacearum*). Primary infection stems from seed-borne inoculum on fuzz/seed coats or infected crop residues. Secondary spread occurs via wind-blown rain, irrigation water, and implements.
  - *Five Symptomatic Phases:*
    1. *Seedling Blight:* Water-soaked, circular-to-irregular dark green spots on cotyledons, causing seedling collapse.
    2. *Angular Leaf Spot:* Small water-soaked, dark green spots on leaf undersides, expanding into angular brown lesions bounded strictly by leaf veinlets. Lesions develop oily margins and dry into necrotic brown spots with yellow halos.
    3. *Vein Blight:* Infection advances along primary and secondary leaf veins, producing black, dead vein streaks.
    4. *Black Arm:* Infection spreads to petioles and main stems, producing elongated, sunken, dark brown-to-black girdling lesions. Stems crack, shred, and break under wind load.
    5. *Bacterial Boll Rot:* Round, water-soaked, dark green depressed spots form on green bolls, turning dark brown/black, causing premature boll drying, seed rot, and lint staining.

- **Alternaria Leaf Spot (*Alternaria macrospora* and *Alternaria alternata*):**
  - *Symptoms: A. macrospora* forms small circular brown necrotic spots (1–10 mm) displaying distinct concentric zonation rings and dry, cracked centers that fall out (shot-hole effect). *A. alternata* forms smaller spots with dark purple margins. Fungi cause severe defoliation and boll rot during heavy boll loading.
- **Grey Mildew / Dahiya Disease (*Ramulariopsis gossypii* / *Ramularia areola*):**
  - *Symptoms:* Angular, pale-green translucent spots (1–10 mm) bounded by veinlets appear on leaves. A white, frosty, powdery fungal growth (conidiophores and conidia) emerges on the abaxial (lower) surface, resembling sprinkled curd or chalk dust. Leaves cup, turn yellowish-brown, dry, and drop prematurely.
- **Target Spot (*Corynespora cassiicola*) and Cercospora Leaf Spot (*Cercospora gossypina*):**
  - *Symptoms: C. cassiicola* forms large circular target-board spots with target-like concentric rings on lower leaves, causing rapid canopy defoliation under high humidity.
- **Fungal Boll Rot Complex:** Caused by *Colletotrichum gossypii* (anthracnose boll rot), *Fusarium* spp., *Aspergillus niger*, and *Rhizopus stolonifer*. Pathogens enter through insect punctures, turning green bolls into dark brown, slimy, rotted masses.

1.5 Soil-Borne Fungal Pathogens and Vascular Wilts

- **Fusarium Wilt (*Fusarium oxysporum* f. sp. *vasinfectum*):**
  - *Pathogen Symptoms:* Soil-borne fungal pathogen surviving via thick-walled chlamydospores. Infects plants at any growth stage. Yellowing (chlorosis) begins along leaf margins on lower leaves, spreading inward along veins, followed by marginal necrosis, leaf wilting, and defoliation from base upward.
  - *Vascular Discoloration Diagnostic:* Longitudinal splitting of main stems or roots exposes dark brown-to-black vascular ring discoloration inside xylem vessels.
- **Fungal Root Rot Complex (*Rhizoctonia solani*, *Rhizoctonia bataticola* / *Macrophomina phaseolina*, and *Sclerotium rolfsii*):**
  - *Symptoms:* Causes sudden wilting, drooping, and complete drying of plants in circular field patches during hot dry spells following irrigation.
  - *Root Diagnostic Identification:* Affected plants are easily uprooted because secondary feeder roots are completely rotted. *R. solani* causes wet brown root rot; *R. bataticola* produces dry, black roots with distinct bark shredding and micro-sclerotia; *S. rolfsii* forms mustard-seed-like spherical sclerotia on root collars.

- **Damping-Off and Seedling Blight Complex (*Pythium* spp. and *Rhizoctonia* spp.):**
  - *Symptoms:* Pre-emergence rot destroys seeds before germination. Post-emergence damping-off forms water-soaked, constricted lesions on succulent hypocotyl stems at the soil line, causing seedlings to topple and die.

1.6 Viral and Phytoplasmal Diseases of Cotton

- **Cotton Leaf Curl Virus Disease (CLCuD):**
  - *Causal Agent and Transmission:* Caused by single-stranded circular DNA virus species belonging to the genus *Begomovirus* (Geminiviridae), associated with DNA-beta and alpha satellite components. Transmitted exclusively by the whitefly vector (*Bemisia tabaci*).
  - *Diagnostic Field Symptoms:* Initial symptoms present as small vein thickening (SVT) on young upper leaves. Progresses into severe upward/downward leaf curling, leaf margin puckering, vein greening, main stem stunting, and formation of cup-shaped leaf-like outgrowths (enations) on abaxial leaf nectaries. Early infection causes total square abortion and yield loss.
- **Tobacco Streak Virus (TSV):**
  - *Causal Agent Vector:* Caused by an *Ilarvirus* transmitted mechanically by thrips (*Thrips tabaci*) carrying virus-contaminated pollen grains.
  - *Symptoms:* Necrotic spots, streaks, and rings on leaves, terminal shoot necrosis, and severe stunting.
- **Cotton Stenosis / Small Leaf Phytoplasma Disease:** Transmitted by leafhoppers, causing proliferation of tiny leaves, witches'-broom growth, floral phyllody, and complete sterility.

1.7 Plant-Parasitic Nematodes in Cotton: Biology and Disease Complexes

- **Root-Knot Nematode (*Meloidogyne incognita*):**
  - *Dominance Life Cycle:* Primary endoparasitic nematode in light, sandy cotton soils. Complete life cycle spans 4 to 6 weeks. Mature females inside roots lay egg masses in a protective jelly-like matrix. The 1st molt occurs inside the egg; the 2nd-instar juvenile (J2) hatches as the sole mobile infective stage. J2 penetrates young root tips, establishes permanent feeding sites, and induces giant cell formation.
  - *Gall Formation Damage:* Triggers hypertrophy and hyperplasia, producing distinct swollen root galls or knots. Disruption of xylem vessels impairs water/nutrient transport, causing plant stunting, leaf chlorosis, uneven field growth, and temporary afternoon wilting during hot summer months.

  - *Nematode-Wilt Disease Complex:* Root damage created by *M. incognita* breaks root epidermal barriers, increasing Fusarium wilt (*F. oxysporum* f. sp. *vasinfectum*) incidence and breaking genetic wilt resistance in cotton cultivars.

- **Reniform Nematode (*Rotylenchulus reniformis*):**
  - *Dominance Diagnostic Symptoms:* Dominant semi-endoparasitic nematode in Central and Southern India, particularly under drip-irrigated monocropping. Does not produce root galls.
  - *"Dirty Root" Symptom:* Infective kidney-shaped female heads enter root cortex while posterior bodies remain outside, secreting gelatinous matrixes that collect fine soil particles, giving roots a characteristic "dirty root" appearance.
  - *Field Decline:* Causes severe root decay, loss of feeder roots, seedling growth cessation at the 3rd leaf stage, stunting, chlorosis, and oval patches of sick crop.

- **Nematode Economic Threshold Level (ETL):** 2 juveniles (J2) per cc or gram of soil.

1.8 Natural Enemies: Predators and Parasitoids in Cotton Ecosystem

- **Predatory Rates of Key Indigenous Defenders:**
  - *Green Lacewing (Chrysoperla carnea):* Each grub consumes up to 100 aphids, 329 whitefly pupae, and 288 jassid nymphs during its developmental cycle.
  - *Ladybird Beetles (Coccinella septempunctata, Cheilomenes sexmaculata):* Adult beetles consume up to 50 aphids or whitefly nymphs per day.
  - *Hoverfly / Syrphid Fly (Syrphus spp.):* 1st instar larva consumes 15–19 aphids/day; 2nd instar consumes 45–52 aphids/day; 3rd instar consumes 80–90 aphids/day (total larval consumption 400 aphids).
  - *Spiders (Lycosidae, Salticidae, Oxyopidae):* Consume 5 large caterpillars or moths per day.
  - *Predatory Mites (Amblyseius spp.):* Adult females consume 20–35 phytophagous spider mites per day.
  - *Generalist Bugs:* Big-eyed bugs (*Geocoris* spp.) and Assassin bugs (*Eocanthecona furcellata*) feed on bollworm eggs and young caterpillars.

- **Key Parasitoids of Cotton Pests:**
  - *Egg Parasitoids: Trichogramma chilonis* and *T. brasiliensis* (parasitizing bollworm eggs; female oviposition capacity 20–200 eggs).
  - *Egg-Larval Larval Parasitoids: Microchelonus versatilis*, *Chelonus blackburni*, *Bracon hebetor* (oviposition capacity 100–200 eggs/female), *Bracon greeni*, *Apanteles angaleti* (pink bollworm larval parasitoid), *Eucarcelia illota* (tachinid larval parasitoid of *H. armigera*), *Eriborus argenteopilosus*, *Campoletis chlorideae*, and *Rogas aligarhensis* (spotted bollworm larval parasitoid).
  - *Whitefly Mealybug Parasitoids: Encarsia formosa*, *Eretmocerus mundus*, and *Aenasius bambawalei* (nymphal parasitoid of *Phenacoccus solenopsis*).

1.9 Agro-Eco System Analysis (AESA) and Field Monitoring

- **Core Components of AESA:**
  1. Plant health and vigor across growth stages.
  2. Built-in compensation abilities of cotton plants (e.g., ability to replace early shed squares).
  3. Pest-to-defender population dynamics and ratios.
  4. Soil moisture and nutrient status.
  5. Microclimatic influences on pest buildup.
  6. Farmer Field School (FFS) experiential learning.
- **Field Scouting Protocol:** Conduct rapid roving surveys at 3 to 5-day intervals across 20 randomly selected plants per acre. Sample 3 leaves per plant (one each from top, middle, lower canopy strata) alongside squares and bolls.
- **Comprehensive Economic Threshold Levels (ETLs) Master Table:**
  - *Cotton Jassid:* $\geq$ 2 nymphs/leaf, or $\geq$ 6 nymphs/3 leaves, or 25% plants displaying Grade-II/III/IV injury.
  - *Cotton Whitefly:* 6 adults/leaf, or 18 adults/3 leaves, or $>$ 50% observed plants showing honeydew/sooty mould.
  - *Cotton Thrips:* $\geq$ 10 thrips/leaf, or 25% plants showing silvery patches on lower leaf surfaces.
  - *Cotton Aphid:* 10% plants showing leaf cupping or crumpling.
  - *Mealybug:* 20 infested plants per acre showing damage Grade-II/III/IV.
  - *Mirid Bug:* 5–10 bugs per 20 plants, or $>$ 5% square damage/staining.
  - *Pink Bollworm:* $>$ 10% rosette flowers (10 out of 100 flowers damaged), or $>$ 10% green boll damage (with live larvae), or pheromone trap catch of 8 moths/trap/night for 3 consecutive nights.
  - *American Spotted Bollworms:* 20% plants showing flared-up squares, or 5% damaged fruiting bodies, or 1 larva/plant, or 4–5 moths/trap/day.
  - *Tobacco Caterpillar (Spodoptera):* 1 skeletonized leaf/plant on 10 plants, or 1 egg mass/20 plants, or 5 larvae/10 plants.
  - *Nematodes (Meloidogyne / Rotylenchulus):* 2 juveniles (J2) per cc or gram of soil.
- **Pheromone and Sticky Trap Infrastructure:**
  - *Yellow Sticky Traps:* Install @ 25 traps/ha (10 traps/acre) at canopy level for whitefly and jassid suppression.
  - *Pheromone Trap Density:* Install funnel traps with specific lures @ 1 trap/ha in North/Central zones and 5 traps/ha in South zone for Pink Bollworm; 5 traps/ha across all zones for *Spodoptera* and *Helicoverpa*; and 2 traps/ha in North zone for *Earias* spp.

1.10 Integrated Cultural, Physical, Biological, and Chemical Management

- **Cultural, Physical, and Mechanical Measures:**
  - *Deep Summer MB Ploughing:* Execute in May–June to invert soil, exposing diapausing bollworm pupae, nematodes, and soil pathogens to solar heat.
  - *Acid Seed Delinting Treatments:* Delint seeds with concentrated H_2SO_4 (100 mL/kg seed). Treat seeds with Imidacloprid 70% WS @ 7 g/kg or Thiamethoxam 70% WS @ 5 g/kg for early sucking pests; Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g/kg or *Trichoderma viride* @ 5–10 g/kg for root rot/wilt; and soak seeds in Streptomycin sulphate + Tetracycline (40–50 ppm) for bacterial blight.
  - *Barrier and Trap Cropping:* Plant 2–3 border rows of maize, sorghum, or pearl millet around cotton fields as windbreaks and predator hubs. Plant border rows of okra or castor as trap crops for *Earias* and *Spodoptera*.
  - *Intercropping:* Intercrop cotton with cowpea, green gram, or groundnut (1:2 ratio) to conserve predators.
  - *Manual Interventions:* Hand-pick and destroy rosette flowers, flared-up squares, and *Spodoptera* egg masses. Detop terminal buds at 100–120 cm height (90–100 DAS).
  - *Post-Harvest Stalk Management:* Allow livestock grazing post-harvest. Shred stalks in-situ with a Super Seeder or stack ex-situ away from fields, treating residue with *Trichoderma harzianum* (@ 5 g/L water) to prevent pathogen carryover.
- **Biological and Biorational Control:**
  - *Parasitoid Releases:* Release *Trichogramma chilonis / T. brasiliensis* @ 1,50000/ha 3–4 times at weekly intervals during early squaring.
  - *Botanicals:* Spray Neem Seed Kernel Extract (NSKE 5%) or Neem Oil 1% (10,000 ppm @ 5 mL/L water) against sucking pests.
  - *Microbial Biopesticides:* Spray *Beauveria bassiana* or *Verticillium lecanii* (1% WP @ 5 g/L) for whiteflies/thrips; *Bacillus thuringiensis* (Bt @ 1.5–2.0 kg/ha) or *HaNPV / SlNPV* (@ 250–500 LE/ha) for caterpillars.
- **Fungicidal Insecticidal Chemical Management (Threshold-Based):**
  - *Sucking Pests:* Flonicamid 50% WG @ 80 g/acre, Diafenthiuron 50% WP @ 200 g/acre, Spiromesifen 22.9% SC @ 200 mL/acre, or Dinotefuran 20% SG @ 60 g/acre.
  - *Bollworms Caterpillars:* Spinetoram 11.7% SC @ 170 mL/acre, Chlorantraniliprole 18.5% SC @ 60 mL/acre, Emamectin benzoate 5% SG @ 80 g/acre, or Indoxacarb 14.5% SC @ 200 mL/acre.
  - *Fungal Diseases:* Foliar spray of Copper oxychloride 50% WP @ 25 g/10L + Streptomycin sulphate (1–2 g/10L) for bacterial blight; Carbendazim 12% + Mancozeb 63% WP @ 30g/10L, Kresoxim-methyl 44.3% SC @ 10 mL/10L, or Azoxystrobin 18.2% + Difenoconazole 11.4% SC @ 10 mL/10L for leaf spots and grey mildew.

1.11 Stage-Wise Integrated Management (IPM/IDM) Seasonal Workflow

1. **Pre-Sowing / Off-Season Window (May–June):**
   - Deep summer MB ploughing; destroy alternate weed hosts (*Parthenium*, *Abutilon*).
   - Select CLCuD-resistant Bt hybrids. Delint seeds and apply chemical/biological seed treatments.
2. **Vegetative Growth Window (0 to 60 Days After Sowing):**
   - Plant 2–3 border rows of maize/sorghum; intercrop with cowpea.
   - Install yellow sticky traps @ 25 traps/ha.
   - Avoid broad-spectrum synthetic pyrethroids to preserve generalist predators (*Chrysoperla*, coccinellids).
   - Apply NSKE 5% or Neem oil 1% if sucking pests cross early ETLs.
3. **Squaring and Flowering Window (60 to 120 Days After Sowing):**
   - Install funnel pheromone traps @ 5 traps/ha for bollworms.
   - Inspect 100 flowers per acre weekly; hand-pick rosette flowers and flared-up squares.
   - Release *Trichogramma* cards @ 1,50000/ha upon initial moth catches.
   - Apply targeted selective chemicals (Flonicamid, Diafenthiuron, Spinetoram, Chlorantraniliprole) or fungicides (Copper oxychloride + Streptomycin, Kresoxim-methyl) only when pests/diseases exceed ETLs.
4. **Boll Maturity and Harvest Window (120 to 180+ Days After Sowing):**
   - Destructively sample 20 green bolls per acre for pink bollworm larvae.
   - Terminate crop canopy by 160–180 DAS. Allow post-harvest livestock grazing, shred stalks in-situ, and apply *Trichoderma* to destroy diapausing pink bollworm larvae and soil pathogens.